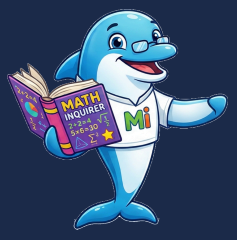
TUTOR GUIDE • LEVEL 1 • TERM 4 • LESSON 9 OF 10

The Problem-Solving Toolbox



Big idea: Pick the tool that fits the problem, then check the answer a second way. We have seven tools: strip, table, list, draw, work backwards, find one first and steps. Look at the problem, pick a tool that fits, solve it, and check the answer another way.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Processes and heuristics (pp. 13 to 14 of the syllabus pages we hold: problem solving, representing, reasoning and communicating; no list of named heuristics). The tools come from Terms 1 to 4 of this program (Choosing a way to solve a one- or two-step problem and saying why (process skills, Primary 1); Strip, table, list, draw it, work backwards, find one first and one step at a time, each with an example (Primary 1 content met in earlier lessons); Guess and check with a table of tries; make equal by sharing out (Primary 1 school papers use both); Beyond Primary 1 on purpose: solve one problem two ways and say why the two answers agree (Two Roads, One Answer), with 3 times as many and a total) Strand label for the tutor only. Every tool in this lesson was met earlier in the year (strips in Lessons 12 and 18, find one first in 20, list in 26 and 36, table in 28, check another way in 29, work backwards and one step at a time in 10, draw it from Lesson 4 on), and the syllabus pages we hold name no list of heuristics, so this lesson teaches the seven tools itself (decision D10): no earlier lesson has a toolbox page. Start here names the tool every time and stays within 100. At Think harder the tool is not named. BEYOND PRIMARY 1 ON PURPOSE at Think harder and Go further only: choosing the tool, a guess-and-check table, make equal two ways, and Two Roads, One Answer: Jo has 3 times as many stickers as Kit and they have 36 in all, solved with a strip and checked with a table of tries (a bar model with times as many is used from Primary 2 and later at school; Lesson 18 met it at Go further, the new part here is checking by a second method). Named puzzle: Two Roads, One Answer. Non-routine types (tutor only): work backwards with the tool not named (T2), draw it (T3), guess and check or a table of tries (T4), make equal (G2), table or work backwards for the day of a pattern (G3). Owner decisions, 22 and 23 Oct 2026: strong content with scaffolding, said plainly to families; every lesson names a beyond item. Do not describe the lesson as MOE-aligned. Wording rule (Term 1 audit, pre-flight D18): wrong working is always a child or a child in another class, never a cast name.
Objectives	• Name the seven tools (strip, table, list, draw, work backwards, find one first, steps) and say when each one helps. • Start here: use the named tool to solve a one- or two-step problem within 100. • Think harder: choose the tool when it is not named, and solve work backwards, draw it and guess-and-check problems. • Go further: solve one problem two ways and say why the answers agree, make three amounts equal, and say which tool you used and why.
Joining cold	**Remember warm-up covers:** Drawing two strips for two numbers and saying how many more (Term 2 Lesson 2, taught again with a picture); Finding the price of 1 from a group of equal things (Term 2 Lesson 10, taught again with a picture); Working backwards from the end to the start on a number line (Term 1 Lesson 10, taught again). Only counting to 20 and adding and taking away small numbers. Nothing from the lessons before is needed: the warm-up re-teaches strips, finding one first and working backwards. The seven tool names are taught in Explore and the Mini-lesson. Not safe to miss: Mi's Challenge Quest in the last lesson of the year has a gate on choosing a tool. A child who misses it watches the recording and does workbook pages 1 and 2. A child who joins here cold can follow: the warm-up re-teaches the three tools a story needs most, and Start here always names the tool.

Child materials	• Your Term 4 book (Book 8), workbook pages 1 to 5, and the tool-and-problem cards page (page 6): the 15 cards (7 tool cards and 8 problem cards) are cut out by a grown-up beforehand • No printer or no cutting? Write the 7 tool names on 7 scraps of paper: strip, table, list, draw it, work backwards, find one first, steps. Your tutor reads the 8 problems aloud • A pencil, a rubber and spare paper
Tutor materials	• The slide deck • Your own set of the 15 cards, if you wish, for yourself (not for the children) • A visible clock Nothing needs to be bought. Cards are small pieces: they stay on the table; nothing goes in the mouth or ears; keep them away from younger brothers and sisters. No matches, toothpicks or anything sharp. Nobody cuts during the lesson: a grown-up cut the cards beforehand.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Beyond Primary 1 (tutor only)	Two Roads, One Answer at G1: Jo has 3 times as many stickers as Kit and they have 36 in all. Road 1, a strip: 4 equal parts, so Kit has 9 and Jo has 27. Road 2, a table of tries: Kit 8 gives 32 in all, Kit 10 gives 40, Kit 9 gives 36. Both roads agree. Choosing the tool with no name (Think harder) and make equal (G2) are the steps before it. Bar models with times as many are used at school from Primary 2 and later; Lesson 18 met 3 times as many at Go further, and the new part here is checking by a second method. Named puzzle: Two Roads, One Answer. Never in Start here, never advertised to children.
Contest-style move	Used at Think harder, Go further and the Investigation: choose the tool that fits (strip, table, list, draw, work backwards, find one first, steps) and say why; work backwards with the tool not named: paid, the change and one price are given, find the other price; draw it: 4th from the front and 3rd from the back, so the line has 6 (the child is counted once); guess and check with a table of tries: 7 coins worth 50 cents, 4 five-cent and 3 ten-cent; make equal: Ali 9, Ben 15, Mei 12, so each ends with 12 and Ben gives 3 to Ali; solve one problem two ways and check that they agree: 3 times as many with a total of 36, by a strip and by a table of tries. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first. Check the answers: S1 7 and 19; S2 18 and 9; S3 15 and 18; S4 25 and 17; S5 5 and 35; S6 15, 25, 30 and 3; T1 Find one first and $25; T2 $9 and $3; T3 3 and 6; T4 4 and 3; G1 9, 27 and Yes; G2 12 and 3; G3 day 7 and an open reason; I1 open. Exit ticket: E1 Work backwards; E2 7 and 18; E3 False.
- Tell parents a week before: the tool-and-problem cards page (page 6 of Book 8) needs 15 cards cut out by a grown-up (7 tool cards and 8 problem cards). The next lesson is the last lesson of the year and holds Mi's Challenge Quest: the Quest Log (4 pages) must be in the house, closed in an envelope; a quest partner (a grown-up or a brother or sister) sits beside the child; and the family finds one favourite page from the year books and plain paper. Send the grown-up note (L40_Quest_Parent_Note.pdf) a week before. Check that every family has Book 8 before the next lesson (it holds the Quest Log): printing and postage are included in the program fee (owner decision, 23 Oct 2026). Parcel policy (draft, owner to confirm): a missing or damaged book is reported, the pages go out as a PDF straight away and there is one free resend; a second loss is discussed. Delivery name, address and phone are deleted 30 days after the parcel is confirmed delivered, or after any resend is finished, whichever is later.
- Have the slides open. Practise the mime-the-tools break (nobody walks backwards). Say the fixed question forms every time: How many more ___ than ___? How many in all? Always say the tool names the same way: strip, table, list, draw it, work backwards, find one first, steps. Never say one word tells the tool: the whole story picks it.
- Check the recording captures the tutor and the shared screen. Children's video is excluded; any unmuted voice is recorded. Greet the whole group first, say the notice, then start the recording; avoid first names while it runs.
- Be plain with families: choosing a tool for a problem is part of the school problem-solving process; choosing the tool when it is not named, make equal, and solving one problem two ways (3 times as many with a total) go beyond Primary 1 on purpose, at Think harder and Go further only. Do not call the lesson MOE-aligned.
- Wording rule: wrong working belongs to 'a child' or 'a child in another class', never to a cast name and never to a child in the group. The stories are made up: no class poll. Slides keep the answers off the screen while you ask: read every question with its numbers aloud. More than one tool can fit most problems: accept a good reason for a different tool.

Safeguarding and recording (this lesson)

- Tool cards and problem cards are small pieces of paper. They stay on the table: nothing goes in the mouth or ears. No matches or toothpicks. Nobody cuts during the lesson; a grown-up cut the cards beforehand.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper, cards or a drawing up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Wrong working belongs to 'a child' or 'a child in another class', never to a child in the group, and never to a name. Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- The stories use made-up names and numbers. There is no class poll and no count by child, so no child is named with a number or a choice.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Nobody walks backwards: the work-backwards move is a slow turn of the arms. A child may stay seated and move one arm.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: The Problem-Solving Toolbox	1:32
2–8	**Remember warm-up**	6	3: Warm-up: draw two strips, Find one first, Work backwards	4:57
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	2:59
12–23	**Explore**	11	6: Explore: your seven tools, Explore: problems 1 to 4, A tool that fits: 1 to 4, Explore: problems 5 to 8, A tool that fits: 5 to 8, What did we notice?	9:13
23–35	**Mini-lesson**	12	9: Seven tools, seven clues, Strip: more and fewer, Rae has 8 cards, 22 in all, Table: a growing pattern, 20 tiles in the 7th path, Work backwards, My number is 15, Two roads, one answer, Road 2: a table of tries	10:06
35–39	**Brain break**	4	1: Mime the tools	3:14
39–70	**Workbook**	31	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	25:14
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:31
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:58
82–85	**Look-back**	3	1: Look-back	2:06
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:34
	Total	**90**		**69:23**

The script needs 69:23 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 81:49. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 39 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 39, skip in this order: Start

with: shorten Explore round 3 (what did we notice) to one question; Then: take the table of tries in the last Mini-lesson example (s-m5-ans) to the answer only; Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:32 of 2:00

0:00	SLIDE	Show slide **The Problem-Solving Toolbox** (s-title)
0:10	DO	Greet the whole group warmly, before the recording starts. Check that you can see each child, or ask the grown-up. [0:12]
0:22	SAY	Hello, inquirers! Pencil ready, and your workbook. Cameras can be on or off, as you like.
0:30	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
0:48	DO	Start the recording now. Do not use children's first names on the recording. [0:05]
0:52	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your tool-and-problem cards. Pause or rewind any time.*
1:04	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera.
1:18	SAY	Today we open the problem-solving toolbox. We meet seven tools, we pick the tool that fits a problem, and we check our answer a second way.
	NOTE	Say the recording statement and the microphone routine in the same words every lesson, for anyone who joined late. Say the notice before you start the recording, and do not use children's first names once it is running. The routine and the four inquirer rules were introduced earlier in the year: do not re-teach them.
	NOTE	Plain statement for families when it comes up: choosing a tool for a problem is part of the school problem-solving process; choosing the tool when it is not named, make equal, and solving one problem two ways go beyond Primary 1 on purpose, at Think harder and Go further only.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. Nothing from earlier lessons is needed.

2:00 – 8:00 Remember warm-up

script 4:57 of 6:00

2:00	SLIDE	Show slide **Warm-up: draw two strips** (s-rem1)
2:12	REC	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*
2:20	SAY	Two strips. Ali has eight stickers and Mei has three.
2:26	ASK	How many more stickers does Ali have than Mei? [wait about 35 s for answers]
3:05	CHILDREN	Your turn. How many stickers do they have in all? Say it with me. [0:30]
3:35	SLIDE	Show slide **Find one first** (s-rem2)
3:47	SAY	Three pens cost twelve dollars. All the pens cost the same. We share twelve into three equal parts.
3:56	ASK	How much does one pen cost? [wait about 35 s for answers]
4:34	CHILDREN	Your turn. One pen costs four dollars. How much do five pens cost? Say it with me. [0:30]
5:04	SLIDE	Show slide **Work backwards** (s-rem3)
5:16	SAY	I think of a number, add five, and get twelve. We start at the end and undo the jump.
5:26	ASK	What is my number? Count back with your fingers. [wait about 35 s for answers]

6:05	CHILDREN	Your turn. I think of a number, take away three, and get nine. What is my number? [0:30]
6:35	CHILDREN	Say each answer with me: five more; eleven; four dollars; twenty dollars; seven; twelve. Then stretch your fingers. [0:22]
	IF STUCK	If a child is unsure about strips, lay 3 counters against 8 counters and count the extra ones. For one pen, share 12 counters into 3 piles one at a time. For working backwards, count back 5 steps on fingers from 12.
	NOTE	Answers: Ali has 5 more than Mei; 11 stickers in all; 1 pen costs $4; 5 pens cost $20; my number is 7. Your turns: 11; $20; 12. Children answer with a whisper, fingers or hands. Nobody writes. Joining cold: all you need is counting to 20 and the pictures.

8:00 – 12:00 Sneak-peek puzzles

script 2:59 of 4:00

8:00	SLIDE	Show slide **Sneak-peek puzzles** (s-sneak)
8:12	REC	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:22	SAY	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:34	READ	Puzzle one. Mei has four stickers. Ali has three more stickers than Mei. How many stickers does Ali have? Write the number.
8:46	CHILDREN	Children write on their own paper. [0:30]
9:16	READ	Puzzle two. Mei and Ali have ten stickers in all. Ali has two more stickers than Mei. How many stickers does Mei have? I know because blank.
9:31	CHILDREN	Children write or draw their answer. Any way is fine: a strip, counters or a guess. [1:20]
10:51	SAY	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: 7 stickers. Puzzle 2 answer: 4 stickers (Mei 4, Ali 6). A child who writes 6 has found Ali: do not correct it now. Look for good ideas to praise later (a strip, a table of tries, counters). Do not write scores or keep a list.

12:00 – 23:00 Explore

script 9:13 of 11:00

12:00	SLIDE	Show slide **Explore: your seven tools** (s-explore1)
12:12	REC	*Recording: pause now. Lay out your 7 tool cards. Put each problem card under a tool you would use. Press play when you have tried.*
12:24	SAY	Today we are toolbox builders. Lay your seven tool cards in a row on the table. A tool is a way to solve a problem.
12:37	READ	Strip. Table. List. Draw it. Work backwards. Find one first. Steps.
12:43	ASK	Which tool have you used before? What did you use it for? [wait about 50 s for answers]
13:39	SLIDE	Show slide **Explore: problems 1 to 4** (s-explore2)
13:51	SAY	Now the problem cards. I read a problem. You put its card under a tool you would use. We do not solve it yet: we only choose.
14:04	READ	Problem one. Ali has nine pens. Mei has four fewer pens than Ali. How many pens do they have in all?
14:16	CHILDREN	Children find card one and lay it under a tool. A volunteer says which tool and why. [0:30]
14:46	READ	Problem two. Rows of blocks have two, four, six blocks, and so on. Each row has two more than the row before. How many blocks are in the sixth row?
15:02	CHILDREN	Children lay card two under a tool. [0:30]
15:32	READ	Problem three. Mei picks two fruits from an apple, a pear and a plum. List every pair she can pick.
15:43	CHILDREN	Children lay card three under a tool. [0:25]
16:08	READ	Problem four. Ravi is third in a line of seven children. How many children are behind Ravi?

16:18	**CHILDREN**	Children lay card four under a tool. [0:25]
16:42	**SLIDE**	Show slide **A tool that fits: 1 to 4** (s-explore2-ans)
16:50	**SAY**	A tool that fits: strip for one, table for two, list for three, draw it for four. More than one tool can fit. A good reason matters more than the name.
17:06	**SLIDE**	Show slide **Explore: problems 5 to 8** (s-explore3)
17:18	**READ**	Problem five. A number take away six is nine. What is the number?
17:25	**CHILDREN**	Children lay card five under a tool. [0:25]
17:50	**READ**	Problem six. Five buns cost fifteen dollars. All the buns cost the same. How much do eight buns cost?
18:00	**CHILDREN**	Children lay card six under a tool. [0:25]
18:25	**READ**	Problem seven. Ben has twenty sweets. He eats eight sweets. Then he buys five more sweets. How many sweets does he have now?
18:38	**CHILDREN**	Children lay card seven under a tool. [0:25]
19:03	**READ**	Problem eight. For every two red beads there are three blue beads. How many blue beads go with eight red beads?
19:14	**CHILDREN**	Children lay card eight under a tool. A volunteer says which tool and why. [0:30]
19:44	**SLIDE**	Show slide **A tool that fits: 5 to 8** (s-explore3-ans)
19:52	**SAY**	Work backwards for five, find one first for six, steps for seven, and a table for eight. Problems two and eight both use a table.
20:05	**SLIDE**	Show slide **What did we notice?** (s-explore4)
20:13	**ASK**	Does one word, like more, tell us the tool? Can two tools fit the same problem? [wait about 45 s for answers]
21:06	**SAY**	The whole story picks the tool, not one word. Next we practise each tool.
	IF STUCK	A child with no cards says the tool name aloud instead. A child who chooses by one word: ask, what does the story give us and what does it ask? A child who is sure about one tool: ask, could another tool fit too?
	NOTE	Main room only. We only choose a tool: do not solve the problems now. Suggested tools: 1 strip; 2 table; 3 list; 4 draw it; 5 work backwards; 6 find one first; 7 steps; 8 table. Other tools may fit: accept a good reason. Do not say the rule before the children do.

23:00 – 35:00 Mini-lesson

script 10:06 of 12:00

23:00	**SLIDE**	Show slide **Seven tools, seven clues** (s-m1)
23:08	**REC**	*Recording: keep watching and copy the examples on your paper. Rewind if you miss a step.*
23:16	**SAY**	One clue for each tool. Strip: when we compare two amounts, or find a total. Table: when the numbers grow in a pattern. List: when we find every way.
23:30	**SAY**	Draw it: for a line, a place or a shape. Work backwards: when we know the end and not the start. Find one first: for equal groups or the price of one. Steps: when one thing happens, then the next.
23:50	**SAY**	A clue is a help, not a rule. Look at the whole story, and check the answer a second way.
24:00	**SLIDE**	Show slide **Strip: more and fewer** (s-m2)
24:08	**SAY**	A strip. Pip has fourteen cards. Rae has six fewer cards than Pip.
24:15	**ASK**	How many cards does Rae have? How many cards do they have in all? [wait about 70 s for answers]
25:32	**SLIDE**	Show slide **Rae has 8 cards, 22 in all** (s-m2-ans)
25:40	**SAY**	Fourteen take away six is eight cards for Rae. Fourteen and eight make twenty-two cards in all. The strip showed us the six fewer.
25:52	**SLIDE**	Show slide **Table: a growing pattern** (s-m3)

26:00	**SAY**	A table. A path has two tiles, then five, then eight. Each path has three more tiles than the path before.
26:10	**ASK**	How many tiles are in the seventh path? Fill in the boxes. [wait about 80 s for answers]
27:36	**SLIDE**	Show slide **20 tiles in the 7th path** (s-m3-ans)
27:44	**SAY**	Two, five, eight, eleven, fourteen, seventeen, twenty. The seventh path has twenty tiles.
27:51	**SLIDE**	Show slide **Work backwards** (s-m4)
27:59	**SAY**	Work backwards. I think of a number. I add nine. Then I take away four. I get twenty.
28:08	**ASK**	What is my number? Undo the last step first. [wait about 85 s for answers]
29:38	**SLIDE**	Show slide **My number is 15** (s-m4-ans)
29:46	**SAY**	Twenty and four make twenty-four. Twenty-four take away nine is fifteen. Check forwards: fifteen and nine make twenty-four, and twenty-four take away four is twenty.
29:58	**SLIDE**	Show slide **Two roads, one answer** (s-m5)
30:06	**SAY**	Two roads, one answer. Pip has twice as many marbles as Rae. They have eighteen marbles in all.
30:15	**ASK**	How many marbles does Rae have? Use the strip. [wait about 90 s for answers]
31:50	**SLIDE**	Show slide **Road 2: a table of tries** (s-m5-ans)
31:58	**SAY**	Road one, the strip: three equal parts, and eighteen shared into three is six. Road two, a table of tries: four is too few, five is too few, six is just right. Both roads say six marbles.
32:16	**ASK**	Why do the two roads agree? [wait about 40 s for answers]
32:59	**SAY**	They use the same story. When two tools agree, we can trust the answer.
	IF STUCK	If a child writes 20 for Rae's cards, say the 6 is how many fewer, not how many cards. If a child writes 21 for the table, check how many steps of 3: from the 1st path to the 7th path there are 6 steps. If a child writes 25 for working backwards, say: undo the add with a take away. If a child writes 9 for Rae's marbles, ask: how many parts does Pip have?
	NOTE	Key words to use every time: tool, strip, table, list, work backwards, find one first, steps, check. Fixed question forms: How many more ___ than ___? How many in all? The slides keep the answers off the screen while you ask: read every question with its numbers aloud. Class numbers are not used in the workbook.

35:00 – 39:00 Brain break

script 3:14 of 4:00

35:00	**SLIDE**	Show slide **Mime the tools** (s-break)
35:08	**REC**	*Recording: stand up and join in! Mime each tool with me, then sit down.*
35:15	**SAY**	Stand up and get a sip of water. We mime the seven tools. Do it with me.
35:24	**CHILDREN**	Mime the tools: draw a long strip in the air, tap a table, write a list in the air, draw a circle, turn your arms backwards like a slow wheel, hold up one finger, climb three small steps with your hands. Then call a tool and let children do its move. Nobody walks backwards. Seated children move one arm. [2:50]

39:00 – 70:00 Workbook

script 25:14 of 31:00

39:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
39:08	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
39:17	**READ**	Page 1, Start here. Every question tells you the tool. Question 1: use a strip. Kumar has twelve marbles. Ravi has five fewer marbles than Kumar. Part a: Ravi has blank marbles. Part b: they have blank marbles in all. Question 2: do one step at a time. A shop has thirty apples. It sells twelve apples in the morning and nine apples in the afternoon. Part a: after the first step, blank apples are left. Part b: after the second step, blank apples are left. Question 3: make a table. Row one has three beads, row two has six beads and row three has nine beads. Each row has three more beads than the row before. Part a: row five has blank beads. Part b: row six has blank beads.

40:28	**CHILDREN**	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [9:00]
49:28	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
49:36	**READ**	Question 1: seven marbles, and nineteen marbles in all. Question 2: eighteen apples, then nine apples. Question 3: fifteen beads, and eighteen beads. Tick your own page with a tick.
49:53	**SAY**	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
50:02	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
50:10	**READ**	Page 2. Question 4: work backwards. Ravi thinks of a number. He takes away eight and gets seventeen. Part a: Ravi's number is blank. Part b: check. Ravi's number take away eight is blank. Question 5: find one first. Four bags hold twenty apples. All the bags hold the same number. Part a: one bag holds blank apples. Part b: seven bags hold blank apples. Question 6: use a list. Ali has three coins: five cents, ten cents and twenty cents. He picks two coins. Part a: five cents and ten cents make blank cents. Part b: five cents and twenty cents make blank cents. Part c: ten cents and twenty cents make blank cents. Part d: there are blank different pairs of coins.
51:17	**CHILDREN**	Page 2, questions 4 to 6. [8:20]
59:37	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
59:45	**READ**	Question 4: twenty-five, and seventeen. Question 5: five apples, and thirty-five apples. Question 6: fifteen, twenty-five and thirty cents, and three pairs. Tick your own page.
60:00	**SLIDE**	Show slide **Choose your next page** (s-choice)
60:08	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
60:24	**MOVE**	Stand and reach up for 30 seconds, then sit down and choose.
60:54	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [3:20]
	IF STUCK	Q1: draw the long strip first, then the shorter one. Q2: do the first sale, then the second. Q3: fill the table box by box, adding 3 each time. Q4: undo the take away with an add. Q5: share the apples into the bags first. Q6: write each pair of coins and add them.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder has a choose-the-tool question, a work-backwards money story, a line with a place from the front and from the back, and a coins puzzle with a table of tries: none of these says which tool except the first. Go further has Two Roads, One Answer (beyond Primary 1 on purpose), make three amounts equal, and a pattern with a sentence frame. Do not announce who is on which page. Start here takes a middle child about 17 minutes, so the choice time may be short. Slow class: take the extra minutes from the flex time and do not skip the answer slides.

70:00 – 76:00 Talk time

script 4:31 of 6:00

70:00	**SLIDE**	Show slide **Talk and explain** (s-talk)
70:08	**REC**	*Recording: pause. Tell a grown-up how you could check an answer a second way.*
70:15	**READ**	A child in another class says: Pip has twice as many marbles as Rae. They have twenty-four marbles in all. Twenty-four shared into two is twelve, so Rae has twelve marbles. Do you agree? How could we check a second way?
70:37	**CHILDREN**	Think about it. You may use your paper. [0:50]
71:27	**CHILDREN**	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]

72:07	**ASK**	Who would like to share? Volunteers first. A pass is always fine. [wait about 100 s for answers]
73:53	**SLIDE**	Show slide **Back to the sneak-peek** (s-sneakans)
74:01	**SAY**	Remember the sneak-peek? Puzzle one: seven stickers. Puzzle two: Mei has four stickers. Many of you used a different tool, and that is good.
74:13	**ASK**	Did anyone change their mind while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: no, we do not agree. Check forwards: if Rae has twelve, Pip has twenty-four, and together they have thirty-six, not twenty-four. Check with a strip: Rae has one part and Pip has two parts, so there are three equal parts, and twenty-four shared into three is eight: Rae has eight and Pip has sixteen. The child shared into two parts but there are three. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:58 of 6:00

76:00	**SLIDE**	Show slide **Exit ticket** (s-exit)
76:08	**REC**	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:16	**SAY**	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:24	**READ**	E1. Mei thinks of a number. She adds four and gets fifteen. Circle the tool that fits best: Table, Work backwards or List. E2. Use a strip. Ali has eleven stickers. Mei has four fewer stickers than Ali. Part a: Mei has blank stickers. Part b: they have blank stickers in all. E3. Five buns cost twenty dollars. A child in another class says one bun costs five dollars. Circle True or False.
77:04	**CHILDREN**	Children answer E1 to E3 on their own. [3:35]
80:39	**SLIDE**	Show slide **Check the exit ticket** (s-exitans)
80:47	**READ**	E1: work backwards. E2: seven stickers, and eighteen stickers in all. E3: false.
80:54	**SAY**	Tick your own ticket with a tick. Well done.
	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result. E3: 20 shared into 5 is 4, so one bun costs $4.

82:00 – 85:00 Look-back

script 2:06 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:08	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:14	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
82:46	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:34 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:08	**SAY**	Well done, inquirers. Next lesson is the last lesson of the year: Inquirers' Showcase and Mi's Challenge Quest. Please have your Quest Log from Book 8, closed in its envelope, a quest partner beside you, a favourite page from your book, and plain paper. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: how many cuts?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time. A rope is cut into five pieces. How many cuts are there? Draw it.
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*

SAY Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).

NOTE This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 4 cuts (5 pieces need 4 cuts). If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Can two different tools solve the same problem? Take problem 4 (Ravi is 3rd in a line of 7): solve it by drawing a line, then by a list of places. Do both give the same answer?	*Recording: pause. Solve a problem with two different tools and see if the answers agree.*	Yes. Drawing a line of 7 children with Ravi 3rd shows 4 behind him. A list of the places 1 to 7 shows places 4, 5, 6 and 7 behind him. Both give 4. A right answer fits every tool.
Make a puzzle for your grown-up	4 (optional)	Pick one tool card. Write a short problem that fits it. Solve it, then ask your grown-up to solve it with the same tool.	*Recording: pause. Pick a tool card, write a problem for it and ask a grown-up to solve it.*	No single answer. Check: the problem has a story, numbers and one question, and the tool fits.
Teach it back	3 (optional)	Tell a toy why we check an answer a second way. Use the Pip and Rae marbles: one road is a strip and one road is a table of tries.	*Recording: pause. Tell a toy why we check an answer a second way.*	If two different tools give the same answer, we can trust it. If they give different answers, one of them has a mistake and we look again.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): Mei has 4 stickers. Ali has 3 more stickers than Mei. How many stickers does Ali have? Write the number.

Answer: 7 stickers.

Puzzle 2 (think): Mei and Ali have 10 stickers in all. Ali has 2 more stickers than Mei. How many stickers does Mei have? I know because ___.

Answer: 4 stickers. Mei 4 and Ali 6 make 10, and 6 is 2 more than 4.

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Use a strip. Kumar has 12 marbles. Ravi has 5 fewer marbles than Kumar.**



(a) Ravi has ☐ marbles.
(b) They have ☐ marbles in all.

Answer: 7 marbles; 19 marbles in all

Hint (say this first): Draw the long strip for Kumar. Ravi's strip is 5 shorter. Then add the two strips for the total.
Worked solution: Ravi: 12 take away 5 is 7 marbles. In all: 12 + 7 = 19 marbles.
Common mistakes: Writing 17 for Ravi (12 and 5 added).; Writing 12 + 5 for the total.
Time for a middle child: about 3 min.

S2 **Do one step at a time. A shop has 30 apples. It sells 12 apples in the morning and 9 apples in the afternoon.**

(a) After the first step, ☐ apples are left.
(b) After the second step, ☐ apples are left.

Answer: 18 apples; 9 apples

Hint (say this first): Do the morning first. Take 12 away from 30. Then do the afternoon.
Worked solution: After the morning: 30 take away 12 is 18 apples. After the afternoon: 18 take away 9 is 9 apples.
Common mistakes: Taking 12 and 9 away from 30 in one go and writing 9 for both parts.; Adding 12 and 9.
Time for a middle child: about 2.5 min.

S3 **Make a table. Each row has 3 more beads than the row before. Fill in the table.**

Row	1	2	3	4	5	6
Beads	3	6	9	☐	☐	☐

(a) Row 5 has ☐ beads.
(b) Row 6 has ☐ beads.

Answer: 15 beads; 18 beads

Hint (say this first): Add 3 each time: 3, 6, 9, then the next box.
Worked solution: 3, 6, 9, 12, 15, 18. Row 5 has 15 beads and row 6 has 18 beads.
Common mistakes: Writing 12 for row 5 (stopping one box early).; Writing 5 times 3 for row 6.
Time for a middle child: about 2.5 min.

S4 **Work backwards. Ravi thinks of a number. He takes away 8 and gets 17.**

(a) Ravi's number is ☐ .
(b) Check: Ravi's number take away 8 is ☐ .

Answer: 25; 17

Hint (say this first): Start at 17 and undo the take away with an add.
Worked solution: 17 and 8 make 25, so Ravi's number is 25. Check: 25 take away 8 is 17.
Common mistakes: Writing 9 (17 take away 8).; Not checking.
Time for a middle child: about 3 min.

S5 **Find one first. 4 bags hold 20 apples. All the bags hold the same number of apples.**



(a) 1 bag holds ☐ apples.
(b) 7 bags hold ☐ apples.

Answer: 5 apples; 35 apples

Hint (say this first): Share 20 apples into 4 equal bags. Then count 7 bags of that number.
Worked solution: 20 shared into 4 bags is 5 apples in 1 bag. 7 bags: 7 groups of 5 is 35 apples.
Common mistakes: Writing 24 (20 + 4).; Writing 7 + 5 for 7 bags.
Time for a middle child: about 3 min.

S6 **Use a list. Ali has three coins: 5¢, 10¢ and 20¢. He picks 2 coins. Fill in the blanks.**



(a) 5¢ and 10¢ make ☐¢.
(b) 5¢ and 20¢ make ☐¢.
(c) 10¢ and 20¢ make ☐¢.
(d) There are ☐ different pairs of coins.

Answer: 15¢, 25¢, 30¢; 3 pairs

Hint (say this first): Write each pair of coins and add them. Pick each pair only once.
Worked solution: 5¢ and 10¢ make 15¢. 5¢ and 20¢ make 25¢. 10¢ and 20¢ make 30¢. There are 3 different pairs.
Common mistakes: Adding all three coins.; Writing 6 pairs (counting 5¢ and 10¢, then 10¢ and 5¢ again).
Time for a middle child: about 3 min.

Think harder (page 3): the taught idea in a new setting

T1 **Read the story. 3 rulers cost $15. All the rulers cost the same. Siti buys 5 rulers.**

(a) Circle the tool that fits best.
(b) 5 rulers cost $☐ .

Answer: Find one first; $25

Hint (say this first): Which tool shares $15 into 3 equal parts? Then count 5 of them.
Worked solution: Find one first: $15 shared into 3 rulers is $5 for 1 ruler. 5 rulers cost 5 groups of $5, which is $25.
Common mistakes: Choosing Table (the number of rulers asked, 5, is not in the pattern 3, 6, 9).; Writing $20 (15 + 5).
Time for a middle child: about 4 min.

T2 **Ben pays with a $10 note for a toy and a card. The toy costs $6. His change is $1.**

(a) Ben spends $☐ in all.
(b) The card costs $☐ .

Answer: $9; $3

Hint (say this first): What did Ben spend in all? Then take the toy away from that.
Worked solution: Ben spent $10 take away $1, which is $9. The toy cost $6, so the card costs $9 take away $6, which is $3. Check: 6 + 3 = 9 and 10 take away 9 is 1.
Common mistakes: Writing $4 (10 take away 6).; Writing $5 (6 and 1 taken from 10 the wrong way).
Time for a middle child: about 5 min.

T3 **Siti is 4th from the front of a line. She is also 3rd from the back. How many children are in the line? Use the box to show your thinking.**

(a) ☐ children are in front of Siti.
(b) There are ☐ children in the line.

Answer: 3 children; 6 children

Hint (say this first): Show the line with a dot for each child. Put Siti in the right place.
Worked solution: There are 3 children in front of Siti. There are 2 children behind Siti, because she is 3rd from the back. 3 + 1 + 2 = 6 children. Siti is counted once.
Common mistakes: Writing 7 (4 + 3, counting Siti twice).; Writing 5.
Time for a middle child: about 5 min.

T4 **Mei has 7 coins. Some are 5¢ coins and some are 10¢ coins. The coins make 50¢ in all. You may use the table.**

10¢ coins	5¢ coins	Total

(a) Mei has ☐ coins of 5¢.
(b) Mei has ☐ coins of 10¢.

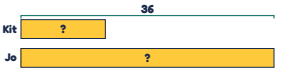
Answer: 4 coins of 5¢; 3 coins of 10¢

Hint (say this first): Try a number of 10¢ coins. Then the rest are 5¢ coins. Is the total 50¢? Try again.
Worked solution: Try 2 coins of 10¢: 5 coins of 5¢ make 25, and 20 + 25 is 45, too little. Try 3 coins of 10¢: 4 coins of 5¢ make 20, and 30 + 20 is 50. So 4 coins of 5¢ and 3 coins of 10¢.
Common mistakes: Guessing without checking the total.; Writing 3 and 4 the wrong way round.
Time for a middle child: about 5 min.

Go further (page 4): mastery and extension

G1 **Jo has 3 times as many stickers as Kit. They have 36 stickers in all. Solve it two ways: a strip and a table of tries.**

Kit	Jo	In all
8	24	32
10	30	40
9	☐	36



(a) Road 1, the strip: Kit has ☐ stickers.
(b) Road 2, the table: when Kit has 9, Jo has ☐ stickers.
(c) Do both roads give the same answer? Circle Yes or No.

Answer: 9 stickers; 27 stickers; Yes

Hint (say this first): The strip has 4 equal parts. In the table, 36 is between 32 and 40. Try Kit 9.
Worked solution: Road 1, the strip: Kit has 1 part and Jo has 3 parts, so there are 4 equal parts. 36 shared into 4 is 9, so Kit has 9 stickers. Road 2, the table: Kit 8 gives 32 in all, and Kit 10 gives 40 in all. Try Kit 9: Jo has 27 and 9 + 27 = 36. Both roads say Kit has 9. Jo has 27.
Common mistakes: Writing 12 for Kit (36 shared into 3 parts).; Trying numbers in a different order and losing track.
Time for a middle child: about 7 min.

G2 **Ali has 9 stickers, Ben has 15 stickers and Mei has 12 stickers. Only Ben and Ali move stickers. Then all three children have the same number.**



(a) Each child ends with ☐ stickers.
(b) Ben gives ☐ stickers to Ali.

Answer: 12 stickers; 3 stickers

Hint (say this first): Add all the stickers. Share them into 3 equal parts. Who has too many? Who has too few?
Worked solution: 9 + 15 + 12 = 36 stickers. 36 shared into 3 children is 12 each. Mei already has 12. Ben has 3 too many and Ali has 3 too few, so Ben gives 3 stickers to Ali.
Common mistakes: Making Ali and Ben equal without looking at Mei.; Writing 6 (the difference between 9 and 15).
Time for a middle child: about 6 min.

G3 **Siti reads 4 pages on day 1, 7 pages on day 2 and 10 pages on day 3. Each day she reads 3 more pages than the day before. Finish the sentence about the tool you used.**

(a) She reads 22 pages on day ☐ .
(b) I used ___ because ___ . (Many answers are right.)

Answer: Day 7; many answers are right

Hint (say this first): A table of days and pages helps. Or start at 22 and work backwards.
Worked solution: A table: 4, 7, 10, 13, 16, 19, 22. She reads 22 pages on day 7. Or work backwards: 22 take away 4 is 18, and 18 is 6 steps of 3, so day 7. Check: 6 steps from day 1 is day 7.
Common mistakes: Writing 6 (the number of steps, not the day).; Not saying why.
Time for a middle child: about 5 min.

Investigation (page 5): Pick a tool

Easy first step: Pick one tool card. Write its name: ____
Then: Write a short problem that fits the tool. Solve it. Then check it a second way. Write your problem and your answer.

> **Model answer: Many answers are right. A good problem has a story, numbers, one question, and an answer that fits the tool. The check uses a second tool or works the problem forwards.**

Look for: A child who writes a short story with numbers and a question, solves it with the chosen tool, and checks it a second way (a different tool, or working forwards). Credit any clear problem. A child who copies a card problem has not yet made their own: ask, what would you change? A child who uses a number that gives a whole answer has thought about the check.
Common mistakes: Writing a problem the tool does not fit.; Writing an answer without a check.

Talk and explain (minutes 70 to 76)

A child in another class says: Pip has twice as many marbles as Rae. They have 24 marbles in all. 24 shared into 2 is 12, so Rae has 12 marbles. Do you agree? How could we check a second way? Sentence starter: I think ... because ...

> **Model answer: No. If Rae has 12, then Pip has twice as many, which is 24, and together they have 36, not 24. A strip has 3 equal parts, and 24 shared into 3 is 8, so Rae has 8 and Pip has 16.**

Look for: A child who checks forwards (12 and 24 make 36, not 24) or who draws the strip with 3 equal parts. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Mei thinks of a number. She adds 4 and gets 15.**
Circle the tool that fits best.

> **Answer: Work backwards**

Worked solution: We know the end (15) and not the start, so we work backwards: 15 take away 4 is 11.
Common mistakes: Choosing Table (no pattern grows).; Choosing List (we do not list every way).

E2 **Use a strip. Ali has 11 stickers. Mei has 4 fewer stickers than Ali.**
(a) Mei has ☐ stickers.
(b) They have ☐ stickers in all.

> **Answer: 7 stickers; 18 stickers**

Worked solution: Mei: 11 take away 4 is 7. In all: 11 + 7 = 18.
Common mistakes: Writing 15 (11 and 4 added).; Leaving out one strip when finding the total.

E3 **5 buns cost $20. A child in another class says 1 bun costs $5.**
Circle True or False.

> **Answer: False**

Worked solution: $20 shared into 5 buns is $4 for 1 bun, not $5. Check: 5 buns at $5 would cost $25.
Common mistakes: Choosing True (using the 5 from 5 buns).; Dividing 20 by 4.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (picking a tool; using a strip; working backwards; something else); *What helped me* (the tool cards; drawing it; checking a second way; I was not stuck); *Next time I will try* (picking a tool; checking a second way; saying why; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Use a strip. Aisha has 15 shells. Kumar has 6 fewer shells than Aisha. How many shells do they have in all? They have ___ shells.	24
H2	Start here	Make a table. Picture 1 uses 4 sticks, picture 2 uses 8 sticks and picture 3 uses 12 sticks. Each picture uses 4 more sticks than the picture before. How many sticks does picture 6 use? ___ sticks.	24
H3	Start here	Work backwards. Lily thinks of a number. She adds 7 and gets 25. Her number is ___ .	18
H4	Think harder	Mei has 5 coins. Some are 10¢ coins and some are 20¢ coins. The coins make 70¢ in all. How many coins of 20¢ does Mei have? ___ coins of 20¢.	2
H5	Start here	A cup hides some counters. The cup and 5 loose counters make 14 counters in all. How many counters are in the cup? ___	9

Online practice (`out/l39/practice`, key `t4l39`): 13 items; 10 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to choose a way to solve a problem and to check it. This week: seven tools, picking the tool, and solving one problem two ways.

Term line (plan, section 5): Your child is learning to choose a way to solve a problem, and to make and test their own puzzles. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**The Problem-Solving Toolbox** s-title	TODAY WE WILL: / Meet seven tools. / Pick the tool that fits. / Check an answer a second way.	Lesson 9 of Term 4, 90 minutes. Objective: name the seven tools (strip, table, list, draw, work backwards, find one first, steps), pick a tool that fits a problem, solve it, and check the answer a second way. Plain statement for families: choosing a tool for a problem is part of the school problem-solving process; choosing the tool when it is not named, make equal, and solving one problem two ways (Two Roads, One Answer) go beyond Primary 1 on purpose, at Think harder and Go further only. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 23, mini-lesson 23 to 35, brain break 35 to 39, workbook 39 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Say the recording notice before you start the recording, in one line: you, the shared screen and any unmuted voice; a child may keep the camera off.
2	2–8	**Warm-up: draw two strips** s-rem1	[picture: bar_model] / Ali has 8 stickers. Mei has 3 stickers. How many more stickers does Ali have than Mei?	Strips (Term 2 Lesson 2, taught again with a picture). The two strips show 8 and 3. The part that sticks out is how many more. Ask: how many more stickers does Ali have than Mei (5). A child who is unsure lays 3 counters against 8 counters and counts the extra ones. Your turn, no new picture: how many stickers do they have in all (11). Children answer aloud. No writing.
3	2–8	**Find one first** s-rem2	[picture: objects] / 3 pens cost $12. All the pens cost the same. How much does 1 pen cost?	Finding the price of 1 (Term 2 Lesson 10, taught again with a picture). 3 pens cost $12 and all cost the same, so share $12 into 3 equal parts: 1 pen costs $4. A child who is unsure shares 12 counters into 3 piles, one counter at a time. Your turn, no new picture: 1 pen costs $4, so how much do 5 pens cost ($20). Children answer aloud. No writing.
4	2–8	**Work backwards** s-rem3	[picture: number_line] / I think of a number. I add 5. I get 12. What is my number?	Working backwards (Term 1 Lesson 10, taught again on a number line). The number line shows a jump of 5 that ends at 12, and the start is an empty box. Ask: what is my number (7). Say: start at the end and undo the jump: 12 take away 5. A child who is unsure counts back 5 steps on fingers from 12. Your turn, no new picture: I think of a number, take away 3, get 9: what is my number (12). Children answer aloud. No writing.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Mei has 4 stickers. / Ali has 3 more stickers than Mei. / How many stickers does Ali have? / PUZZLE 2: think: / Mei and Ali have 10 stickers in all. / Ali has 2 more stickers than Mei. / How many stickers does Mei have? / I know because ___.	Puzzle 1 is an access question: Ali has 4 and 3 more, so 7 stickers. Puzzle 2 is a thinking question on the lesson idea: Mei and Ali have 10 in all and Ali has 2 more, so Mei has 4 and Ali has 6. Children may use any way: a strip, guess and check, a table of tries, or counters. A child who writes 6 has found Ali, not Mei: that is a good start, do not correct it now. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: 7 stickers; 4 stickers. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.

6	12–23	**Explore: your seven tools** s-explore1	[picture: table] / Lay your 7 tool cards in a row. Which tool have you used before?	Round 1, about 2 minutes. Children lay the 7 tool cards in a row on the table (a grown-up cut them out beforehand). Read the seven names: strip, table, list, draw it, work backwards, find one first, steps. Ask: which tool have you used before, and what did you use it for? Take two or three answers from volunteers; a pass is fine. Say: a tool is a way to solve a problem. Today we practise picking one. Nobody holds a card up to the camera: children keep the cards on the table. A child with no cards writes the seven names on scraps of paper.
7	12–23	**Explore: problems 1 to 4** s-explore2	Card | Problem / 1 | Ali has 9 pens. Mei has 4 fewer pens than Ali. How many pens do they have in all? / 2 | Rows of blocks have 2, 4, 6 blocks, and so on. Each row has 2 more than the row before. How many blocks are in the 6th row? / 3 | Mei picks 2 fruits from an apple, a pear and a plum. List every pair she can pick. / 4 | Ravi is 3rd in a line of 7 children. How many children are behind Ravi? / Put each problem card under a tool you would use. Say why.	Round 2, about 4 minutes. Read problem 1 aloud with its numbers, then give 20 seconds for children to find card 1 and lay it under a tool card. Ask one or two volunteers: which tool, and why. Then problem 2, 3 and 4 the same way. There is no need to solve the problems: we only choose a tool. Answers (a tool that fits) are on the next slide. Say: more than one tool can fit; the reason matters more than the name. A child who chooses by one word ("more" so add) has not yet looked at the whole story: ask, what do we know and what do we want to find? Nobody holds a card up to the camera.
8	12–23	**A tool that fits: 1 to 4** s-explore2-ans	Card | A tool that fits / 1 | Strip / 2 | Table / 3 | List / 4 | Draw it / More than one tool can fit. A good reason matters more than the name.	The answers for round 2. Problem 1: strip (two strips, one 4 shorter). Problem 2: table (the rows grow by 2 each time). Problem 3: list (every pair, none missed). Problem 4: draw it (a line of 7 children). Other tools can fit too: problem 4 can also be a list. Read it aloud so that children can tick their own cards with a tick. Never name a child who chose a different tool: a different tool with a good reason is a good answer.
9	12–23	**Explore: problems 5 to 8** s-explore3	Card | Problem / 5 | A number take away 6 is 9. What is the number? / 6 | 5 buns cost $15. All the buns cost the same. How much do 8 buns cost? / 7 | Ben has 20 sweets. He eats 8 sweets. Then he buys 5 more sweets. How many sweets does he have now? / 8 | For every 2 red beads there are 3 blue beads. How many blue beads go with 8 red beads? / Put each problem card under a tool you would use. Say why.	Round 3, about 3 minutes. The same routine as round 2 with problems 5 to 8: read each problem aloud with its numbers, 15 seconds to find the card and lay it down, then one volunteer says which tool and why. Two of the problems (2 and 8) share the table tool. Answers (a tool that fits) are on the next slide. A child who puts every problem under strip has found one tool they like: ask what the story gives and what it asks.
10	12–23	**A tool that fits: 5 to 8** s-explore3-ans	Card | A tool that fits / 5 | Work backwards / 6 | Find one first / 7 | Steps / 8 | Table / Problems 2 and 8 both use a table. Many tools can fit one problem.	The answers for round 3. Problem 5: work backwards (start at the end and undo). Problem 6: find one first (5 buns cost $15, so 1 bun costs $3, then 8 buns). Problem 7: steps (take away 8, then add 5). Problem 8: table (2 red with 3 blue, 4 red with 6 blue, 6 red with 9 blue, 8 red with 12 blue). Problems 2 and 8 share the table tool. Say: the tool is chosen by what the story gives us, not by one word. Children tick their own cards with a tick. Do not name any child.
11	12–23	**What did we notice?** s-explore4	Does one word, like more, tell us the tool? / Can two tools fit the same problem?	Take one or two answers from volunteers. Key ideas: a word like more does not choose the tool, the whole story does; more than one tool can fit the same problem; a tool is not a trick, it is a way to see the problem; after solving, we can check with another tool. Repeat each child answer in your own words so the recording picks it up. Do not give the rule before the children say it; the mini-lesson makes it exact.

12	23–35	**Seven tools, seven clues** s-m1	[picture: table]	One clue for each tool, read aloud in the order of the table. Strip: compare two amounts, or find a total. Table: the numbers grow in a pattern. List: find every way. Draw it: a line, a place or a shape. Work backwards: you know the end, not the start. Find one first: equal groups, or the price of one. Steps: one thing happens, then the next. Say: a clue is a help, not a rule: look at the whole story. Say: after the answer, check it a second way. About 90 seconds.
13	23–35	**Strip: more and fewer** s-m2	[picture: bar_model] / Pip has 14 cards. Rae has 6 fewer cards than Pip. How many cards do they have in all?	Worked example, strip. Pip has 14 cards. Rae has 6 fewer cards than Pip. Children work out how many cards Rae has and then how many cards in all, on their own paper. Ask: how many cards does Rae have (8); how many in all (22). The answer is on the next slide. Say: draw the long strip first, then a shorter one, and the gap is the 6 fewer. A child who answers 20 has added 14 and 6: the 6 is how many fewer, not how many cards. Check: 14 take away 8 is 6.
14	23–35	**Rae has 8 cards, 22 in all** s-m2-ans	[picture: bar_model] / 14 take away 6 is 8 cards for Rae. 14 + 8 = 22 cards in all.	The answer for the example on the slide before. Rae: 14 take away 6 is 8. In all: 14 and 8 make 22. Read it aloud so that children can tick their own work. Say: the strip showed us the 6 fewer, so we knew to take 6 away.
15	23–35	**Table: a growing pattern** s-m3	[picture: table] / A path has 2 tiles, then 5 tiles, then 8 tiles. Each path has 3 more tiles than the path before. How many tiles are in the 7th path?	Worked example, table. The tiles grow by 3 each time: 2, 5, 8. Children fill the empty boxes on their own paper: 11, 14, 17, 20. Ask: how many tiles are in the 7th path (20). The answer is on the next slide. Say: a table shows how the numbers grow. A child who writes 21 has added 3 one time too many: the 4th path is 11. Check: 20 take away 2 is 18, and 6 steps of 3 are 18.
16	23–35	**20 tiles in the 7th path** s-m3-ans	[picture: table] / Add 3 each time: 2, 5, 8, 11, 14, 17, 20. The 7th path has 20 tiles.	The answer for the example on the slide before. 2, 5, 8, 11, 14, 17, 20. The 7th path has 20 tiles. Read it aloud so that children can tick their own work. Say: from the 1st path to the 7th path there are 6 steps of 3, and 6 threes are 18.
17	23–35	**Work backwards** s-m4	[picture: table] / I think of a number. I add 9. Then I take away 4. I get 20. What is my number?	Worked example, work backwards. The boxes show the story: start, add 9, a middle number, take away 4, the end 20. Children work back from the end on their own paper: undo the take away 4 (20 and 4 make 24), then undo the add 9 (24 take away 9 is 15). Ask: what is my number (15). The answer is on the next slide. Say: undo the last step first. A child who writes 25 has added 9 again instead of taking it away.
18	23–35	**My number is 15** s-m4-ans	[picture: table] / Start at 20 and undo each step: 20 + 4 = 24, then 24 take away 9 is 15. Check: 15 + 9 = 24 and 24 take away 4 is 20.	The answer for the example on the slide before. Undo the last step first: 20 and 4 make 24. Then undo the first step: 24 take away 9 is 15. Check forwards: 15 and 9 make 24, and 24 take away 4 is 20. Read it aloud so that children can tick their own work. Say: checking forwards is a second way.
19	23–35	**Two roads, one answer** s-m5	[picture: bar_model] / Pip has twice as many marbles as Rae. They have 18 marbles in all. How many marbles does Rae have?	Worked example, two roads. Pip has twice as many marbles as Rae and they have 18 marbles in all. Road 1: the strip. Rae has 1 part and Pip has 2 parts, so there are 3 equal parts, and 18 shared into 3 parts is 6 marbles for Rae. Children work out how many marbles Rae has on their own paper. Ask: how many marbles does Rae have (6). The answer is on the next slide, with a second road: a table of tries. Say: twice as many is a Primary 2 and later idea at school; we try it because the strip shows it. A child who writes 9 has shared 18 into 2 equal parts: ask, how many parts does Pip have?

20	23–35	**Road 2: a table of tries** s-m5-ans	[picture: table] / Road 1, the strip: 3 equal parts, 18 shared into 3 is 6. Road 2, the table: Rae 6 and Pip 12 make 18. Both roads say Rae has 6 marbles.	The answer for the example on the slide before. Road 1, the strip: 3 equal parts, so Rae has 6 and Pip has 12. Road 2, a table of tries: try Rae 4 (Pip 8, in all 12, too few), Rae 5 (Pip 10, in all 15, still too few), Rae 6 (Pip 12, in all 18, just right). Both roads agree. Ask: why do the two roads agree (they use the same story, so a right answer must fit every part of it). Say: when two tools agree, we can trust the answer. A child who tries numbers at random is guessing; a table of tries goes up in order and gets closer.
21	35–39	**Mime the tools** s-break	Stand up. Mime each tool with me. / Draw a long strip in the air. Tap a table. / Sip some water.	Four minutes. Stand up on camera with a little clear space. Mime the tools together: strip (draw a long line in the air with both hands), table (tap a table with a finger), list (write a list in the air), draw it (draw a circle), work backwards (turn your arms backwards, like a slow wheel; nobody walks backwards), find one first (hold up one finger), steps (climb three small steps with your hands). Then say a tool name and let children do its move. A child may stay seated and move one arm. No equipment, no running. Do not give prizes or compare children.
22	39–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Take your time. / questions S1, S2, S3	Read questions 1 to 3 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. Question 1: use a strip: Ravi has 5 fewer marbles than Kumar, and how many in all. Question 2: do one step at a time: 30 apples and two sales. Question 3: make a table for a pattern of beads. Every Start here question names its tool.
23	39–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1 Ravi has 7 marbles and 19 in all. Q2 18 apples left after the first step and 9 after the second. Q3 15 beads in row 5 and 18 beads in row 6.
24	39–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. Take your time. / questions S4, S5, S6	Read questions 4 to 6 aloud. Question 4: work backwards: Ravi takes away 8 and gets 17, so his number, and a check. Question 5: find one first: 4 bags hold 20 apples, so 1 bag, then 7 bags. Question 6: use a list: three coins, pick two, three pairs, how many pairs. Wait 10 seconds before helping.
25	39–70	**Check page 2** s-ws2ans	answers S4, S5, S6	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q4 Ravi's number is 25 and the check is 17. Q5 1 bag holds 5 apples and 7 bags hold 35 apples. Q6 15, 25 and 30 cents, and 3 different pairs.
26	39–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | The tool is not named: you choose, and say why / Go further (page 4) | Two Roads, One Answer, make three amounts equal, a pattern / Investigation (page 5) | Pick a tool, write a problem / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.

27	70–76	**Talk and explain** s-talk	A CHILD IN ANOTHER CLASS SAYS: / Pip has twice as many marbles as Rae. / They have 24 marbles in all. / 24 shared into 2 is 12, so Rae has 12 marbles. / Do you agree? How could we check a second way? / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no, we do not agree. Check forwards: if Rae has 12 then Pip has twice as many, 24, and together they have 36, not 24. Check with a strip: Rae has 1 part and Pip has 2 parts, so there are 3 equal parts, and 24 shared into 3 is 8. Rae has 8 and Pip has 16. The child shared into 2 parts but there are 3. Link to the mini-lesson: two roads, one answer. Repeat each child answer in your own words for the recording. The child in the story is in another class: no cast name is given the wrong answer.
28	70–76	**Back to the sneak-peek** s-sneakans	Puzzle 1: 7 stickers. Puzzle 2: Mei has 4 stickers. Mei 4 and Ali 6 make 10, and 6 is 2 more than 4.	Reveal both puzzles now. Say: many of you wrote a good answer, and many of you used a different tool. Do not read out anyone earlier answer. Puzzle 1: 3 more than 4 is 7. Puzzle 2: a strip, a table of tries (Mei 3 and Ali 5 make 8, too few; Mei 4 and Ali 6 make 10) or counters all give 4. It was the same idea as the mini-lesson: more than one tool can fit, and we can check a second way.
29	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud, with the numbers. Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
30	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: E1 Work backwards; E2 7 stickers and 18 stickers in all; E3 False.
31	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.
32	85–90	**Well done, inquirers!** s-end	Show a grown-up your tool cards. Say which tool you like best. / NEXT LESSON: / Inquirers' Showcase and Mi's Challenge Quest / Have ready: your Quest Log, closed in its envelope, a quest partner, a favourite page from your book and plain paper.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that the next lesson is the last lesson of the year and holds Mi's Challenge Quest: the Quest Log (4 pages) is in Book 8, kept closed in an envelope until the tutor says to open it; a quest partner is a grown-up or a brother or sister; a favourite page from the year book and plain paper are for the showcase; and the tutor sends one line about what we saw in the Quest, never a mark. If the whole class is ahead, show the optional slide and use the Spare-time menu from the tutor guide.
33	optional	**Go further: how many cuts?** s-opt	A rope is cut into 5 pieces. How many cuts are there? Draw it.	Optional, only if you are on time. Draw the rope as a line and cut it: 1 cut makes 2 pieces, 2 cuts make 3 pieces, so 5 pieces need 4 cuts (the last piece needs no cut). Take two answers and move on. This is the draw-it tool again. Say: when we draw it, the answer shows.
34	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Can two different tools solve the same problem? Try one. / Make a puzzle | Pick a tool card and write a problem for your grown-up. / Teach it back | Tell a toy why we check an answer a second way. / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your tool-and-problem cards. Pause or rewind any time.*	Gather your workbook, pencil, paper and your 15 cards. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*	Play along. Say each answer out loud before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Lay out your 7 tool cards. Put each problem card under a tool you would use. Press play when you have tried.*	Pause. Lay out the tool cards. Put each problem card under a tool that fits.
23:00	Mini-lesson	*Recording: keep watching and copy the examples on your paper. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
35:00	Brain break	*Recording: stand up and join in! Mime each tool with me, then sit down.*	Stand up. Mime the seven tools with the movements, then sit down.
39:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up how you could check an answer a second way.*	Pause. Explain your answer to someone at home and show how you know.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up your tool cards and say which tool you like best.*	Show a grown-up your tool cards and say which tool you like best.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:32	1:43	ok
remember	6:00	4:57	5:53	ok
sneak	4:00	2:59	3:29	ok
explore	11:00	9:13	10:47	ok
mini	12:00	10:06	11:53	ok
break	4:00	3:14	3:51	ok
work	31:00	25:14	29:56	ok
talk	6:00	4:31	5:22	ok
exit	6:00	4:58	5:51	ok

look	3:00	2:06	2:27	ok
flex	5:00	0:34	0:38	ok
Total	90:00	69:23	81:49	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 8.0 min for page 1 and 9.0 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.